



Sustainable agriculture creates management trade-offs, not conflicts, between crop productivity and soil carbon storage goals

David G. Encarnation^{a,b,*}, Robert S. Powell^{a,b}, Pete Smith^c, Adam F.A. Pellegrini^{a,b,d}

^a Department of Plant Sciences, University of Cambridge, Cambridge, UK

^b Conservation Research Institute, University of Cambridge, Cambridge, UK

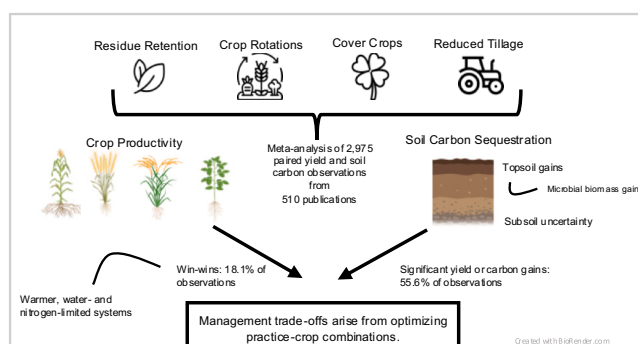
^c Institute of Biological and Environmental Sciences, University of Aberdeen, Cruickshank Building, St Machar Drive, Aberdeen, AB24 3UU, UK,

^d Department of Earth System Science, Doerr School of Sustainability, Stanford University, Stanford, CA, USA

HIGHLIGHTS

- Sustainable agricultural practices can increase soil carbon stocks and crop yields.
- Trade-offs arise because different practices maximize outcomes in different crops.
- Yield and soil carbon gains are more likely in water- and nitrogen-limited areas.
- Uncertainties in subsoil carbon responses may offset topsoil carbon gains.
- Sustainable agricultural practices could be successful across much of the world.

GRAPHICAL ABSTRACT



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ABSTRACT

Accelerating climate change and increasing food insecurity necessitates agricultural management that simultaneously mitigates climate change and increases food production. We evaluated when and where sustainable agriculture creates win-wins or trade-offs between food production and soil carbon sequestration across four crop types and four broad management changes. We conducted a global meta-analysis of 2975 paired yield and topsoil carbon (and 498 subsoil) observations from 510 publications to assess the simultaneous impacts of reduced tillage, cover cropping, complex crop rotations, and crop residue retention on maize, wheat, rice, and soybean yields and soil organic carbon stocks. Strong coupling between crop yield and soil carbon gains occurred in only 18.1% of observations, but significant gains in either outcome occurred in 55.6%. Environmental conditions mattered across all practices and crop types: win-wins were most likely in marginal agricultural lands characterized by warmer temperatures, water limitation, and low nitrogen application rates – conditions that represent a substantial proportion of global agricultural land area. By considering the interactions between crop types and multiple practices thought to bolster soil carbon stocks and crop yields, we conclude that the optimal practice for maximizing the yield response is not always the same as the optimal practice for maximizing the soil carbon response. This generates management trade-offs not via declines in either yields or soil carbon, but rather in determining the optimal practice selection for a given crop in a given area.

* Corresponding author at: Department of Plant Sciences, University of Cambridge, Cambridge, UK.

E-mail address: dge26@cam.ac.uk (D.G. Encarnation).

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1. Introduction

Modified agricultural practices have the potential to both mitigate emissions from the food system (which accounts for a third of anthropogenic emissions) (Crippa et al., 2021) and improve the food supply for over 2 billion people facing food insecurity (FAO, 2024). Given that land available for agriculture is limited, a key opportunity is addressing both issues by adopting practices that simultaneously mitigate climate change and increase food production. One pathway is increasing soil organic carbon (SOC) (P. Smith, 2012), which is often correlated with greater yields of several staple crops (Ma et al., 2023; Oldfield et al., 2019). Increasing soil carbon sequestration and protection may contribute 47% of the climate mitigation potential in grasslands and agriculture (Bossio et al., 2020), though estimates of soil carbon sequestration rates in croplands range widely from 0.25 to 6.78 Gt CO₂-eq yr⁻¹ (Roe et al., 2019). Thus, there is scope to achieve both goals of increasing food production and sequestering carbon simultaneously.

Agricultural practices thought to improve SOC and yield include reduced tillage, cover cropping, complex crop rotations, and improved crop residue management, among others (Paustian et al., 2019; Rehberger et al., 2023). Often included under the umbrella of conservation agriculture (Giller et al., 2015), adopting these practices can affect crop yields and soil carbon sequestration, in both positive (e.g., Bai et al., 2019; Lessmann et al., 2022; Xia et al., 2018) and negative (D. B. Lobell et al., 2025; Ogle et al., 2019; Pittelkow et al., 2015) ways. This creates opportunities to identify win-win scenarios for food production and climate change mitigation but also requires an understanding of where there are trade-offs. Indeed, recent modelling work has suggested that these trade-offs may be more common than previously thought (McClelland et al., 2025). Accordingly, relating soil carbon and yield responses to changes in agricultural management remains one of the main barriers in estimating the climate mitigation potential and scaling implementation of sustainable practices (Amelung et al., 2020; Oldfield et al., 2024). But empirical studies investigating the simultaneous impacts of these practices on soil carbon and crop yields have been either region-specific (e.g., Alvarez et al., 2017) or focused on individual practices (Vendig et al., 2023). Comparing effects across multiple practices and crop types is crucial to maximize soil carbon and crop yield gains by identifying optimal practices. Combining multiple practices may help to alleviate trade-offs (McClelland et al., 2025; Pittelkow et al., 2015) or increase the gains in yields and/or soil carbon stocks (Qiu et al., 2024), though they may also reduce benefits relative to the individual practices (e.g., Prairie et al., 2023).

The magnitude and distribution of win-wins and trade-offs also depend on agronomic conditions and environmental constraints. While there is substantial heterogeneity in the relationships between yields and SOC responses and agronomic and environmental gradients, some trends have been identified. Most of the practices lead to greater soil carbon changes in warmer conditions with faster decomposition of carbon inputs from residue retention or cover crops to form soil organic matter, or where reduced tillage counteracts this enhanced decomposition (Poeplau and Don, 2015; Prairie et al., 2023; Shang et al., 2021). In drier, water-limited regions, practices like reduced tillage and residue retention may help conserve soil moisture, leading to greater gains in crop yields (Pittelkow et al., 2015) and/or soil carbon (Sun et al., 2020). However, cover crops can compete with cash crops for limited water availability, constraining their effectiveness in arid regions (Abdalla et al., 2019). Similarly, soil properties can influence crop productivity and place constraints on SOC accrual. Practices like residue retention and reduced tillage may be more likely to improve yields in sandier soils (Xia et al., 2018) that are water-limited, and SOC benefits tend to be greater in soils with low initial SOC concentrations (Liu et al., 2014; Rehberger et al., 2023). Recent work suggests that the saturation of mineral-associated organic carbon (MAOC) impacts the accrual and vulnerability of SOC stocks and offers a more refined estimate of SOC sequestration potential (Georgiou et al., 2025). Lastly, agronomic

decisions like fertilizer use can produce different responses: cover cropping and complex crop rotations tend to increase yields more in low-N systems (Abdalla et al., 2019; King and Blesh, 2018), whereas residue retention may have a greater effect on crop yields in high-N conditions (Xia et al., 2018). It is also expected that the SOC benefits from these practices take time to accumulate (Liu et al., 2014; Poeplau and Don, 2015; Prairie et al., 2023) but may saturate (Shang et al., 2021). While the effects of these environmental and agronomic gradients on crop yields and soil carbon outcomes are well-documented (often with competing trends), there has not been an assessment of multiple sustainable practices in multiple crops on combined yield and SOC changes that would facilitate practice comparisons aimed at identifying optimal practices.

Moreover, previous research on the effects of these practices on soil carbon sequestration have often been limited to the topsoil (King et al., 2024; McClelland et al., 2021; Sun et al., 2020). Studies that have investigated subsoils have shown both neutral (Prairie et al., 2023; Qiu et al., 2024) and negative (e.g., Ogle et al., 2019) effects, which has thus far limited the scientific consensus around whether subsoil carbon responses may offset topsoil SOC increases (Oldfield et al., 2024; Paustian et al., 2016). Similarly, limited global-scale research suggests that practices such as residue retention and reduced tillage may increase microbial biomass carbon (Li et al., 2018). The response of microbial biomass is one mechanism behind changes in SOC and can inform our understanding of how sustainable agricultural practices affect soil organic matter formation and stabilization (Cotrufo et al., 2013).

We used a global meta-analysis to investigate the potential of four agricultural practices – improved crop residue management, reduced tillage, cover cropping, and complex crop rotations – to sequester carbon in the soil while maintaining or increasing yields of four staple crops: maize, wheat, rice, and soybeans (which together accounted for 46.7% of global cropland area in 2023) (FAO, 2024).

2. Materials and methods

2.1. Data collection

2.1.1. Literature search

We comprehensively searched Web of Science and SCOPUS for studies that quantified both the yield and soil carbon impacts of four agricultural practices: cover cropping, complex crop rotations, reduced tillage, and crop residue retention (Table S1). We searched for studies that investigated the impact of any of these practices on soil carbon concentrations or stocks and yields of four main staple crops (maize, wheat, rice, and soybeans) (Lists S1 & S2). The search included all English-language primary articles published until November 17, 2022. After duplicates were removed, this yielded 13,945 unique articles. We present a detailed description in the Supplementary Material of study selection criteria.

2.1.2. Data compilation & management

In total, we identified 510 papers published between 1978 and 2024 that met our inclusion criteria (Fig. S1), resulting in 2975 paired yield and aggregated topsoil SOC observations from 402 unique study sites. The practices investigated are defined in Table S1. While cover cropping and crop rotations can be distinguished based on the timing of planting (cover crops are usually grown during fallow periods) and the end use of the crop (cover crops usually are not harvested as a cash crop), in practice this distinction relies on information rarely reported in studies. As such, we often combined the two practices into a binary variable: cropping system intensification (“Intensification”), following Prairie et al. (2023).

The full study inclusion criteria are included in the Supplementary Materials. In short, we included observations from studies that compared a single sustainable practice (e.g., reduced tillage) or combination of practices (e.g., reduced tillage plus cover cropping) to the lack

of that practice (control). Apart from the sustainable practice manipulation, the control and treatment were managed the same way (irrigation, fertilization, etc.). Crucially, we required that the study reported both crop yield data as well as soil carbon data for both the control and the treatment plots on similar time scales. If multiple publications reported results from the same study site (e.g., at different times in a long-term experiment, or different trials at the same site), we coded them as the same study. If the study reported different crops from the same treatments (e.g., reported yields for both rice and wheat in a rice-wheat rotation), we included each crop independently (Zhao et al., 2022). We did not differentiate between different seasons of crops (e.g., spring wheat vs winter wheat; early vs late-season rice) as papers did not reliably report this information. Irrigation was coded as a binary variable, and studies that did not report irrigation were labelled as rainfed.

For each paired comparison, we collected the means, number of replications and estimates of variability (standard deviation, standard error, coefficient of variation) for the SOC and yield parameters in conventional and sustainable plots. We imputed missing standard deviations using the average coefficient of variation for all observations with reported standard deviations (Table S2) (Furukawa et al., 2006; Pellegrini et al., 2023). The average study length across all observations was ~7 years, meaning that yield and SOC changes represent the difference in yields between conventional control and sustainable treatment after an average of 7 years.

As we wanted to compare soil carbon stocks, we occasionally imputed missing bulk density values. Instead of using global gridded datasets, which would give the same values for conventional and sustainable groups, we used separate control and treatment mean bulk densities to account for the effects of sustainable practices on bulk density (Al-Kaisi and Lal, 2020). We imputed missing bulk densities separately for topsoil and subsoil observations to preserve the effects of depth on bulk density (Panagos et al., 2024). We used the mean values from the 31.0% of topsoil observations and 40.1% of subsoil observations with reported bulk densities to impute missing values (Table S2). We used the overall treatment and control mean bulk density values rather than treatment-specific means because practice-specific means differed minimally from global means (Fig. S2), some practice categories had very sparse data, and we wanted to avoid overly fine stratification that can actually capture noise from small sample sizes (Deeks et al., 2024; Rothman and Mosquin, 2013) (See Supplementary Material for greater detail).

We also aggregated soil carbon stocks across depth using bulk densities to first calculate stocks in each layer (Prairie et al., 2023). We used a threshold of 30 cm to separate between topsoil (0–30 cm) and subsoil (>30 cm) layers (see Supplementary Material for details on overlapping layer depths). This resulted in 2975 aggregated topsoil observations, 498 aggregated subsoil observations, and 531 aggregated full profile observations, with maximum depths in the full profile ranging from 35 to 105 cm. The “full profile” was defined as any depth-aggregated observation where the maximum depth exceeded 30 cm (i.e., where the soil sampling included both topsoil and subsoil increments). The average maximum topsoil depths were 22.7 cm, 20.6 cm, 19.2 cm, and 18.2 cm for the sole manipulations of reduced tillage, residue retention, cover cropping, and crop rotations, respectively. We also collected 326 aggregated topsoil observations of microbial biomass carbon concentrations. We also used mass-weighted averaging to aggregate soil properties (clay content, pH, etc.) that were reported in studies, using bulk density values from HWSD (FAO and IIASA, 2023) as a neutral, spatially explicit reference. However, the choice of bulk density values had little effect on the depth-aggregated soil properties because most soil property values did not vary by depth (i.e., they were reported at the site level) (Fig. S3) (see Supplementary Materials for more detail).

Environmental and agronomic covariate data were either taken from the papers or supplemented with global gridded datasets (Table S3), using latitude and longitude provided by the study or based on the nearest landmark (e.g., city). The degree of saturation of mineral-

associated organic carbon was derived from Georgiou et al. (2022) and refers to the realized amount of mineral-associated organic carbon (MOC) divided by the potential amount of mineral-associated carbon (MOC_{max}).

2.2. Meta-analysis

2.2.1. Effect size calculation

All statistics were run in R v4.3.2 (R Core Team, 2023). The natural log of the response ratio (lnRR) was used as the effect size to measure the effect of the sustainable agricultural practices on soil carbon stocks and crop yields (Eq. 1).

$$\ln RR_{site} = \ln \left(\frac{\mu_{sustainable}}{\mu_{conventional}} \right) \quad (1)$$

The lnRR was calculated for each paired observation, where the numerator is the mean yield or SOC stock of the ‘sustainable’ practice, and the denominator is mean of the ‘conventional’ practice. Thus, a value >0 illustrates a gain in SOC or yield in the sustainable treatment relative to the conventional control which serves as the baseline. Effect sizes were calculated with the *escalc* function from the *metafor* package (Viechtbauer, 2010). We weighted observations by the inverse of the variance. SOC and yield response ratio outliers were identified based on having a z-score > 3 and were removed before analysis. We transformed response ratios into percentage change for visualization.

2.2.2. Univariate mixed effects models

To visualize how the soil carbon and yield responses were distributed across different thresholds of environmental and agronomic variables, we fitted univariate mixed effects models using the *rma.mv* function in the *metafor* package. This approach allowed us to account for potential non-independencies of data by including study sites as a random effect. For these models, we either used the complete topsoil SOC and yield dataset or practice-specific subsets where observations were only included in a practice-specific subset if that practice was the sole manipulation (i.e., these subsets did not include practice stacking). This was done to isolate the effects of within-practice differences (such as comparing minimum tillage to no-tillage) while minimizing the confounding effects of other practices.

2.2.3. Variable importance

To identify important predictors of soil carbon and yield responses, we first removed highly correlated continuous variables (Pearson correlation coefficient ≥ 0.70). We then used the R package *metaforest* (Lissa, 2020) to fit meta-regression models, which are robust to overfitting and non-linear relationships and incorporate meta-analytical variance and weights.

We performed recursive variable pre-selection for both the complete dataset and the inclusive practice-specific data subsets in *metaforest* with 10,000 iterations and 100 replications using the *preselect* function, retaining variables that improved model performance in >90% of iterations. We then optimized the tuning hyperparameters (Table S4) using the *train* function within the *caret* package (Kuhn, 2008) and selected top models to minimize root-mean-square error (RMSE). Important predictors were identified using an importance threshold of 0.7 in each top model, similar to many previous studies (Du et al., 2020; Pellegrini et al., 2023; Terrer et al., 2021). For the variable importance and meta-regressions (see Section 2.2.4), we included observations in a practice subset if that practice was one of the manipulations, not just the sole manipulation. This allowed us to increase the sample size while controlling for the potentially confounding effects of practice stacking by including practice stacking as a predictor in the model selection process. (see Supplementary Material for details on subset definitions).

2.2.4. Meta-regressions

After establishing variable importance using *metaforest*, we visualized the individual effects of important predictors on topsoil SOC and yield responses using meta-regression analysis. Variables were selected based on their importance in the individual practice-specific yield or SOC models. For visualization purposes, however, variables important for either SOC or yields were included in the final models to illustrate the different effects of a variable on SOC and yields (Table S10, Table S11, Table S12). For each continuous predictor variable, we used the *predict* function from the *metafor* package to generate prediction curves by varying the focal predictor across its observed range while holding all other important variables constant at their mean values (or mode for categorical variables). This allowed for interpretation of the effects of individual predictors while accounting for other important predictors.

2.2.5. Model assumptions

We assessed the presence of publication bias using funnel plots, Egger's regression tests (Egger et al., 1997), the trim-and-fill method (Duval and Tweedie, 2000), and Fail-Safe N Analysis (Rosenberg, 2005). See Supplementary Material for more detail (Table S5, Table S6, Table S7).

3. Results

The complete dataset spanned 402 study sites spanning all six continents with arable agriculture, with 2975 observations of paired yield and topsoil organic carbon stock measurements (Fig. S6). The

representativeness of the dataset in terms of global distributions of important environmental covariates is shown in Figs. S7 and S8.

3.1. Practices improve both yields and topsoil soil carbon stocks globally

Overall, the practices were beneficial for topsoil carbon and crop yields. Win-wins—significant increases in both yield and topsoil SOC—were present in 18.1% of observations, and roughly twice the number of observations (37.5%) contained significant gains in either one of SOC stocks or crop yields under improved management. In contrast, it was rare (5.7% of observations) to experience significant losses of SOC or yield, and exceptionally rare (1% of observations) to experience losses in both soil carbon and yields (Fig. 1). Observations without significant effects totalled 37.8%. Thus, while there was substantial variability in the magnitude of effects, the majority (55.6% of observations) demonstrated significant gains in yield and/or topsoil SOC relative to the conventional controls, while significant losses occurred in only 6.7% of observations.

3.2. Magnitude of topsoil carbon and yield responses depend on crop type and type of practice implemented

Across all four staple crops investigated, sustainable agricultural practices consistently increased mean topsoil SOC stocks by approximately +10.0% (Fig. 2a). The yield response was more variable. Significant yield increases were observed in maize and wheat systems (average increase of +13.1% and + 8.2%, respectively). Soybean yields

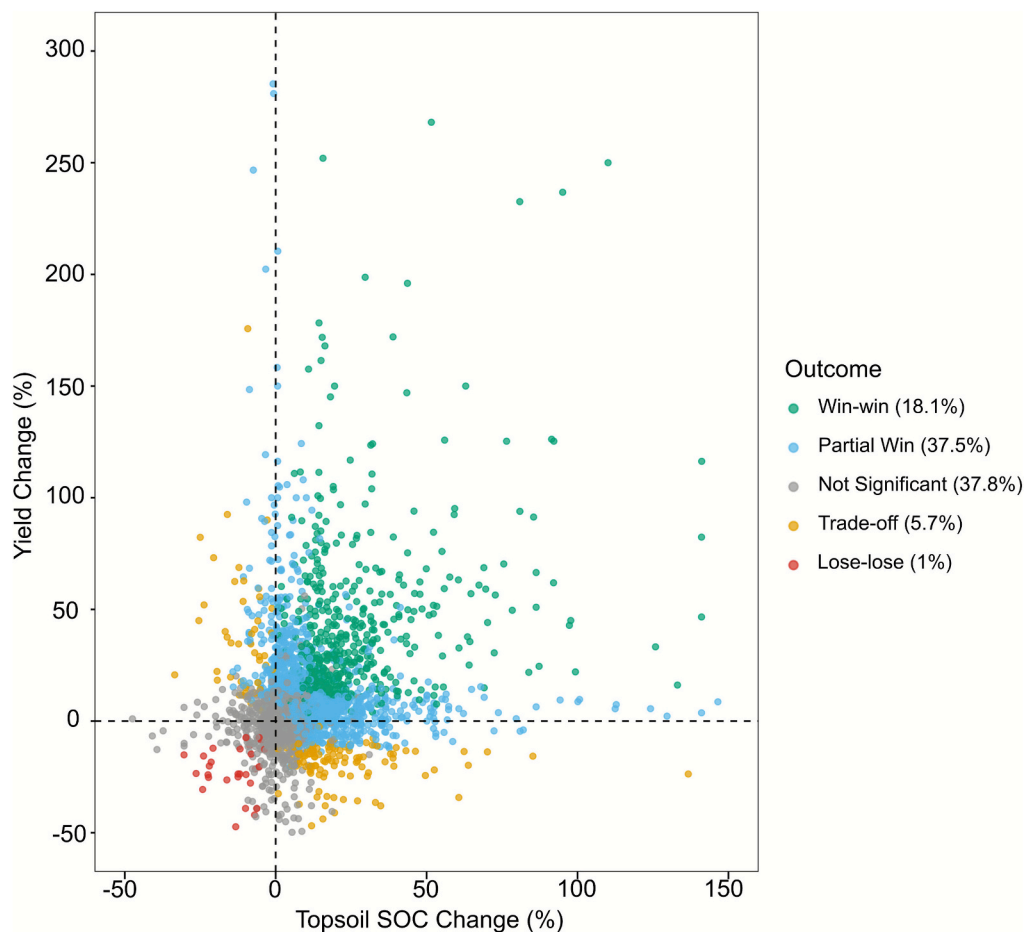


Fig. 1. Distribution of yield and topsoil soil carbon stock responses. Points illustrate the percentage change in crop yields and topsoil SOC for each aggregated topsoil observation. Colours indicate the significance based on the 95% confidence interval of each point estimate (estimated using $1.96 \times \text{standard error}$): Win-win = both responses significantly positive. Partial Win = One response significantly positive while the other response is neutral. Not Significant = both responses non-significant. Trade-off = one response is significantly negative. Lose-lose = both responses significantly negative.

increased by an average of +12.9%, but this was not significant, nor was the +5.7% average increase in rice yields (Table S15). Thus, maize and wheat exhibit more consistent gains in yields while all exhibit consistent gains in topsoil SOC.

Of the individual practices investigated in isolation, reduced tillage by itself led to sizeable topsoil SOC gains (+7.9%) yet insignificant yield increases. On the other hand, intensification (cover cropping and/or crop rotations) by itself led to large average yield increases (+13.1%) but insignificant SOC increases (Fig. 2b). Within intensification, cover cropping alone slightly increased SOC (+6.7%) and led to a large significant increase in crop yields (+26.6%), while crop rotations alone slightly increased SOC (+3.3%) and had no effect on crop yields (+0.4%, $p > 0.5$) (Fig. 2c). Residue retention by itself produced large gains in both SOC (+12.7%) and yields (+11.2%) (Fig. 2b) (Table S15). Thus, while all individual practices are beneficial for at least one outcome, management trade-offs may arise given the variability of SOC and yield responses to different practices.

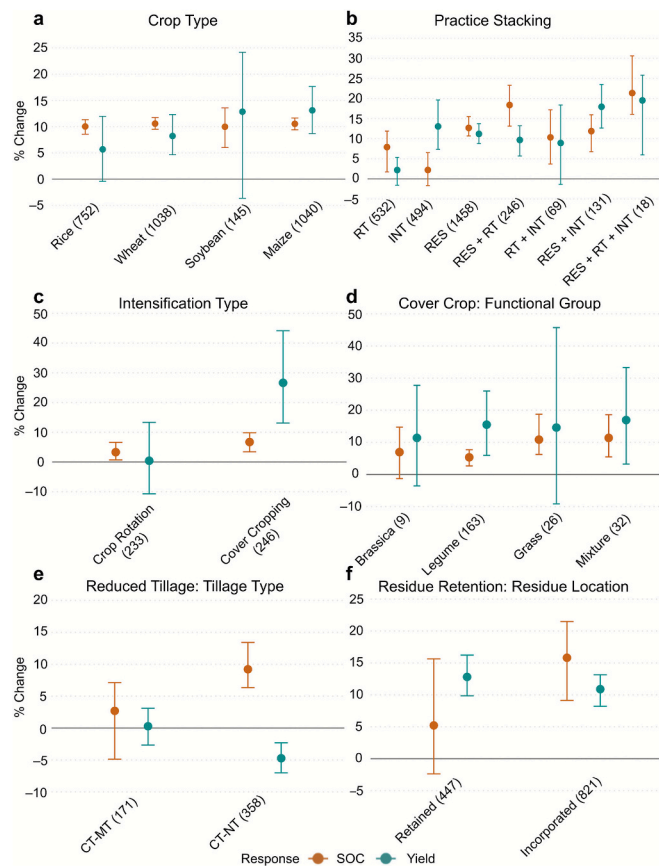


Fig. 2. Overall topsoil SOC and crop yield responses. Broken down by crop type (a) and combination of practices used (b). a and b use the full dataset and c-f use practice-specific datasets for cover cropping (c), crop rotations (d), reduced tillage (e), and residue retention (f). All analyses based on univariate mixed effects models. Points represent mean estimates and error bars represent bootstrapped 95% confidence intervals, such that if the bars do not overlap the x-axis then the effects are significant. Values in parentheses represent sample sizes. In b, RT = reduced tillage, INT = cropping system intensification (cover cropping and/or crop rotations) and RES = residue retention. In c-f, practice-specific data subsets were used, which only include observations where that practice is the sole manipulation (i.e., there is no practice stacking. See Supplementary Material for more details of practice-specific subsets and bootstrapping procedures. CT-MT refers to observations where the conventional practice is conventional tillage and the sustainable practice is minimum tillage; in CT-NT the sustainable practice is no-tillage.

3.3. Practice implementation methods influenced the magnitude of topsoil soil carbon and yield responses

Each individual practice has multiple methods of implementation, which were important for predicting topsoil SOC and crop yield responses. For example, the implementation of reduced tillage can be broadly divided into minimum tillage and no-tillage methods (Table S1), which affected the magnitude and significance of SOC and yield changes: Switching from conventional to no-tillage led to a significant SOC increase (+9.2%), whereas the conversion to minimum tillage did not significantly affect SOC stocks. Switching to minimum tillage had no significant effect on crop yields, but no-tillage significantly reduced yields (−4.7%) (Fig. 2e) (Table S15).

Cropping system intensification can be partitioned into cover cropping and crop rotation methods (Fig. 2c), as well as the species used for cover cropping. The SOC response was consistently positive in all cover crop functional groups except *Brassica* spp. and was slightly higher when using grasses (+10.8%) compared to legumes (+5.4%). Conversely, using a legume cover crop led to a significantly positive yield response (+15.5%) whereas using a grass cover crop species did not significantly affect crop yields. Using a mixture of cover crop functional groups led to large and significant increases in both SOC (+11.4%) and crop yields (+16.9%) (Fig. 2d) (Table S15).

Residue retention effects depended on the depth of addition: the SOC response was only significantly positive when crop residues were incorporated into the soil (+15.8%); surface retention did not significantly affect topsoil SOC stocks. Despite this, both residue incorporation and surface retention led to similar significant yield increases (+10.9% and +12.8%, respectively) (Fig. 2f) (Table S15).

Thus, even within a practice, the method of implementation can change not only the magnitude and significance of yield and topsoil SOC changes, but whether the practice generates co-benefits between SOC and yield.

3.4. Stacking practices enhanced gains in topsoil SOC and yields

Combining multiple practices can have synergistic effects that often exceeded the benefits of individual practices. Combining other practices with reduced tillage tended to increase the average topsoil SOC response (from +12.7% to +18.4% for residue retention and from +2.2% to +10.3% for intensification) though it did slightly decrease the average yield response (from +11.2% to +9.7% for residue retention and from +13.1% to +8.9% for intensification). Combining other practices with intensification tended to increase the average yield response (from +11.2% to +17.9% for residue retention and from +2.2% to +8.9% for reduced tillage, though the latter two were not statistically significant). Residue retention was the most broadly beneficial practice and combining other practices with residue retention increased both SOC and yield responses. This may have helped overcome the negligible impacts of intensification on SOC (increased the average response from +2.2% to +11.9%) and reduced tillage on yields (increased the average response from +2.2% to +9.7%). Lastly, though the sample size was small ($n = 18$), combining all three practices led to the largest mean increases in SOC (+20.4%) and crop yields (+19.5%) (Fig. 2b) (Table S15). Thus, stacking multiple practices can improve outcomes and alleviate neutral or negative impacts of individual practices.

3.5. Efficacy of practices varied by crop type

Crop-practice interactions were important determinants of the topsoil SOC and yield responses. In rice systems, all individual practices increased topsoil SOC, though the average response was largest with residue retention (+11.3%), followed by reduced tillage (+9.9%) and intensification (+6.2%). Intensification and residue retention significantly increased rice yields (+15.4% and +8.7%, respectively), and reduced tillage did not significantly affect yields. In soybean systems,

residue retention increased topsoil SOC (+9.9%) but reduced tillage and intensification did not have significant effects. Conversely, intensification increased soybean yields (+13.0%) but reduced tillage and residue retention did not. In wheat systems, all practices increased topsoil SOC, though the average response was largest with residue retention (+12.8%), followed by reduced tillage (+8.8%) and intensification (+5.3%). Residue retention also increased maize yields (+9.7%), but reduced tillage and intensification did not have significant effects. Lastly, in maize systems, residue retention significantly increased topsoil SOC (+13.8%) and yields (+15.2%). Intensification significantly increased maize yields (+13.1%) but had no effect on SOC, and reduced tillage did not significantly impact either outcome (Table 1). These findings demonstrate that the effect of the practices may vary by crop type, but there is not always a clear 'best practice' for a given crop.

3.6. SOC and yield gains were greater on marginal land with historically less intensive management

Several climatic, soil, and farm management gradients were significantly related to topsoil SOC and yield responses. For climate, the greatest SOC response was observed in arid regions (+17.7%), though it was consistently positive across humid, dry sub-humid, and semi-arid zones (+9.6%, +12.6%, and + 9.4%, respectively). Conversely, the lowest yield response was observed in arid regions (+4.2%) relative to wetter zones (+8.6%, +11.3% and + 10.6%, in humid, dry sub-humid, and semi-arid regions, respectively) (Fig. 3a) (Table S15). Thus, while SOC and yields may increase across all aridity zones, more arid regions are less likely to experience yield gains despite the large increase in SOC stocks.

For soil properties, the greatest SOC response was observed in soils with low initial (background) SOC contents (+12.5%), followed by medium- and high-SOC content soils (+10.2% and + 9%, respectively). But the background SOC content had a variable impact on the yield response, with the greatest yield response in high-SOC soils (+12.9%) and the lowest yield response in medium-SOC soils (+7.7%) (Fig. 3b). Similarly, the average SOC response was greatest (albeit marginally) in the soils with the lowest level of mineral-associated organic carbon (MAOC) saturation (+11%) whereas the average yield response was greatest in the soils with the highest level of MAOC saturation (+10.5%) (Fig. 3c) (Table S15). Thus, soil carbon gains tend to be larger in soils with low initial soil carbon concentrations, while the trend with MAOC saturation is more variable.

Nitrogen application rate did not have a clear effect on the SOC response, with average responses ranging from +8.7% in the 160–240 kg N/ha category to 11.1% in the >240 kg N/ha category. There was, however, a clear negative relationship between nitrogen application rate and yield response. The greatest yield response was observed in

unfertilized systems (+16.3%), followed by low-N systems (<160 kg N/ha; +10.6%), and finally medium-N (+6%) and high-N systems (+6.6%) (Fig. 3d) (Table S15). While SOC relationships were reasonably consistent across crop types, crop-specific patterns revealed important nuances in the relationship between N rate and yield response (Table S9). For maize, the yield response was greatest in unfertilized systems (+23.6%) and declined as the N rate increased, reaching +10.6% at >240 kg N/ha. Wheat showed a similar pattern, with the yield response becoming insignificant at high N rates. In rice, the yield response was also greatest in unfertilized systems (+19.4%) and lower in high N systems (+9.6%), though the relationship was non-linear, with insignificant yield changes in intermediate N rates. Lastly, soybean yields responded differently than the cereals: the yield responses were similar in unfertilized (+15.0%) and high-N systems (+21.3%), with insufficient data for intermediate N ranges. Thus, nitrogen application tends to reduce the relative gains in yield for cereal crops (but not soybeans) while the gains in SOC are generally consistent across different fertilization levels.

Lastly, both SOC and yield responses accumulated over time. The greatest SOC response was in long-term studies (+15%; >10 years), followed by medium-term (+12.3%; 5–10 years) and smallest in short-term studies (+7.4%; <5 years). Similarly, the yield response was greatest in long-term and medium-term studies (+10.9% and + 10.4%, respectively) and smallest in short-term studies (+8.4%) (Fig. 3e) (Table S15). Thus, both SOC and yield responses may be maximized over longer time periods, but the effect of time may have a stronger impact on the SOC response.

3.7. SOC and yield responses across practices were greater in stressful and low N environments

The effects of reduced tillage on both topsoil SOC and yield responses were primarily influenced by climatic variables (Table S10) and were positively associated with mean annual temperature (Fig. 4a). Both topsoil SOC and yield responses to reduced tillage were negatively associated with potential evapotranspiration (Fig. 4b) and aridity index (aridity index is the inverse i.e., more arid regions have a lower aridity index) (Fig. 4c), though the relationships with SOC response were only weakly significant (Table S12). The mean yield response became negative in areas with lower mean annual temperature (<10° C) and in humid climates (aridity index >0.65), whereas the SOC response was consistently positive and changed minimally across the gradients. Thus, reduced tillage maximizes the combination of SOC and yield responses in warmer and drier conditions.

The effects of intensification practices (cover cropping and crop rotations) were influenced primarily by farm management variables such as crop type and N application rate, and to a lesser extent by climatic

Table 1
Effects of selected practices broken down by crop type. Mean values represent % change (converted from response ratio) of crop yields and topsoil SOC stocks for each combination of crop type and agricultural practice. 95% confidence intervals and *p*-values derived from 1000-iteration bootstrapping, and *n* reflects the sample size in each combination of crop and practice. Practice subsets refer to the sole manipulations. Intensification = just cover cropping and/or complex crop rotations. See Table S7 for full breakdown including practice stacking.

Crop type	Practice	Crop yields			Topsoil SOC stocks			<i>n</i>
		Mean	95% CI	<i>P</i> value	Mean	95% CI	<i>P</i> value	
Rice	Reduced tillage	−0.1	−9.2, 7.3	0.968	9.9	6.9, 13.0	<0.001	95
	Intensification	15.4	8.7, 22.2	<0.001	6.2	1.6, 10.3	0.018	170
	Residue retention	8.7	3.0, 14.1	<0.001	11.3	8.8, 13.8	<0.001	357
Wheat	Reduced tillage	0.2	−4.7, 6.0	0.970	8.8	5.5, 11.2	<0.001	200
	Intensification	9.6	−1.9, 20.3	0.092	5.3	2.0, 10.5	<0.001	133
	Residue Retention	9.7	6.7, 12.8	<0.001	12.8	10.6, 15.7	<0.001	528
Soybean	Reduced tillage	3.0	−16.3, 18.1	0.662	7.1	−0.6, 12.9	0.067	47
	Intensification	13.0	1.1, 30.5	0.032	2.5	−5.9, 11.7	0.625	39
	Residue Retention	12.3	−25.7, 45.4	0.41	9.9	3.5, 15.7	<0.001	36
Maize	Reduced tillage	4.1	−3.9, 10.1	0.262	7.6	−1.0, 12.9	0.102	190
	Intensification	13.1	0.9, 26.8	0.032	−3.2	−9.5, 4.0	0.348	152
	Residue Retention	15.2	10.1, 20.0	<0.001	13.8	11.1, 17.9	<0.001	537

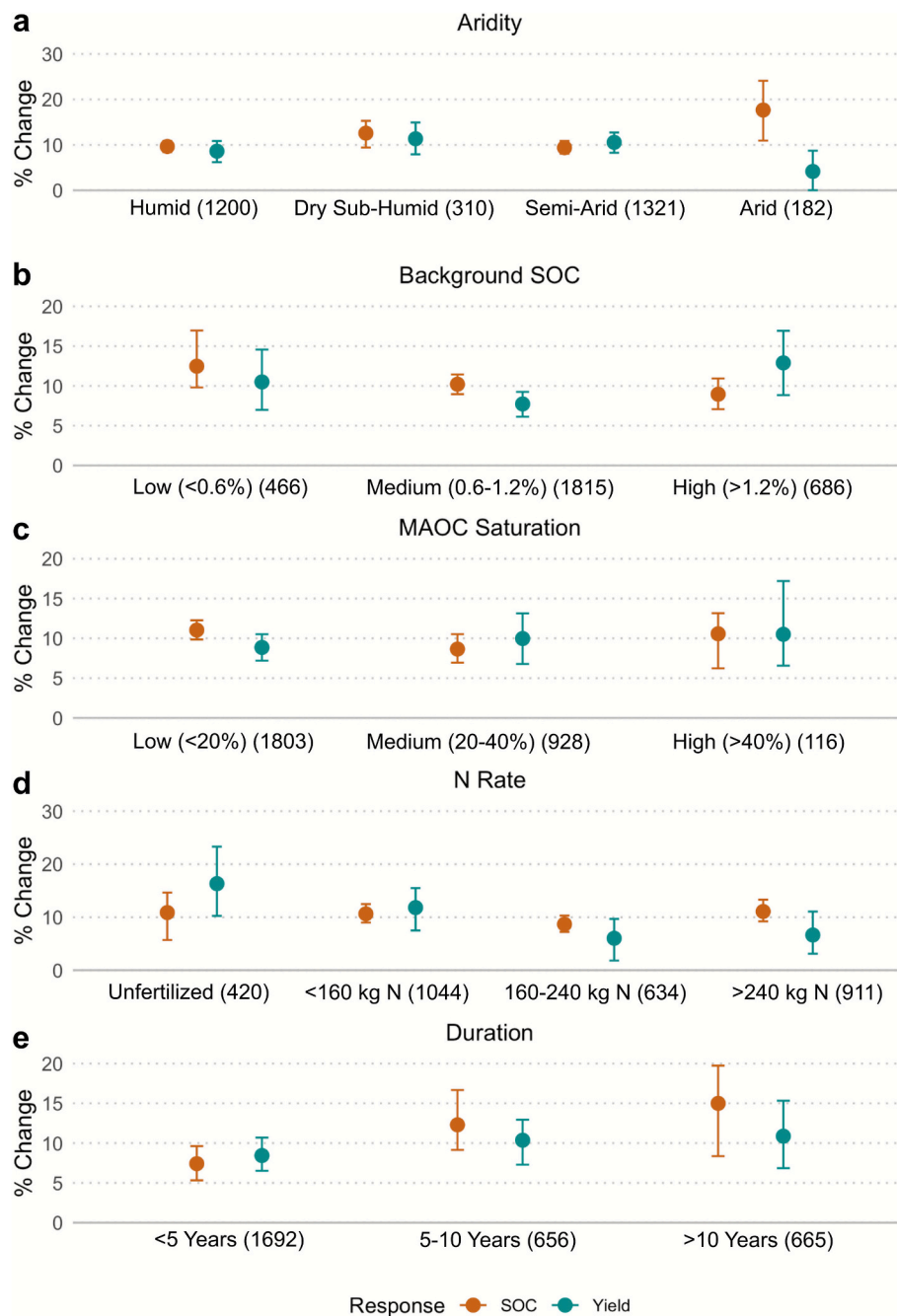


Fig. 3. Overall topsoil SOC and crop yield responses broken down by selected soil, climate and management predictors. All analyses based on univariate mixed effects models. Points represent mean estimates and error bars represent bootstrapped 95% confidence intervals, such that if bars do not overlap the x-axis then the effects are significant. Values in parentheses represent sample sizes. Aridity index (AI) derived from Zomer et al. (2022). Arid and hyper arid: $0 < AI < 0.2$; Semi-Arid: $0.2 \leq AI < 0.5$; Dry sub-humid: $0.5 \leq AI < 0.65$; Humid: $AI \geq 0.65$. Background SOC derived from HWSO (FAO and IIASA, 2023). MAOC Saturation is derived from Georgiou et al. (2022) and refers to the realized amount of mineral-associated organic carbon (MOC) divided by the potential amount of mineral-associated carbon (MOC_{max}). Nitrogen application rate and study duration derived from the studies themselves.

conditions (Table S8). For cover cropping, topsoil SOC and yield responses were positively associated with mean annual temperature (Fig. 4g) (Table S11). Unlike reduced tillage, the yield response to cover cropping was consistently positive, and the SOC response only became significantly positive in warmer climates ($MAT > 20^{\circ}C$).

The yield responses to both cover cropping and crop rotations declined as the nitrogen application rate increased (Figs. 4h,i) (Table S11). With cover cropping, the yield response decreased by ~1.7% for every 10 kg increase in nitrogen application rate, though the thresholds at which the yield response became positive varied by crop

type (Fig. S9). Rice and maize yields increased when nitrogen application was below 160 kg N/ha, wheat yields increased with nitrogen application less than 100 kg N/ha, and soybean yields never significantly increased across the entire nitrogen gradient.

The effects of residue retention were influenced by both climatic and management variables (Table S8). The yield response increased with increasing mean annual temperature (Fig. 4d) though the SOC response was unaffected (Table S12). The yield response correlated negatively but the SOC response correlated positively with potential evapotranspiration (Fig. 4e). Both SOC and yield responses were positively associated

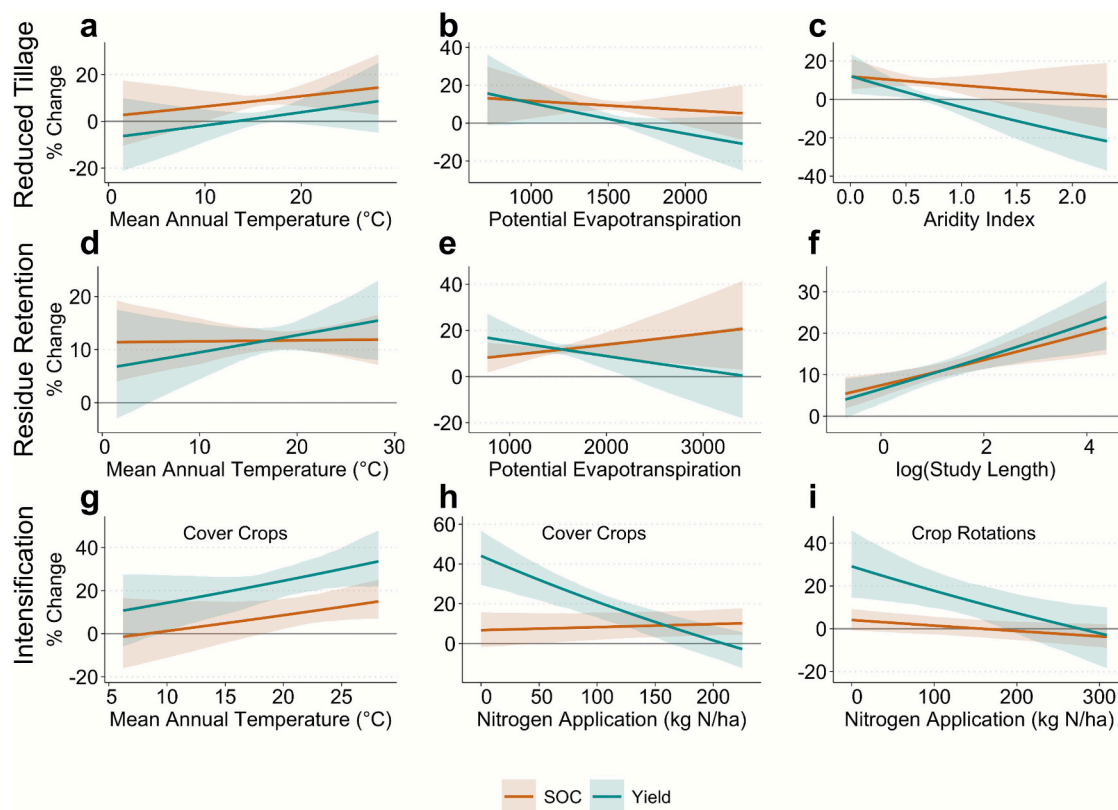


Fig. 4. Practice-specific partial regressions. Models derived from important predictors from practice-specific SOC and yield sub-models, using inclusive data subsetting where observations were retained in a practice subset if that practice was one of the manipulations, not necessarily the sole manipulation (see Supplementary Materials for more detail). Partial regressions generated by varying one predictor while holding others at their means and categorical variables at their most common level when applicable. Cover cropping plots (4 g & 4 h) are set to the most common crop type (Rice) and crop rotation plot (4i) is set to the most common management level (Intensification alone). Prediction curves based on 100 equally spaced model predictions (see Supplementary Material). Error bars represent bootstrapped 95% confidence intervals, such that if bars do not overlap the x-axis then the effects are significant. Note that aridity index is the inverse of aridity i.e., more arid regions have a lower aridity index.

with study duration (Fig. 4f). Thus, the benefits of residue retention take time to accrue despite some climatically controlled trade-offs when trying to maximize for either SOC or yields.

Taken together across all practices, these results suggest that sustainable agricultural practices are more likely to produce win-wins in stressful environments such as warmer, more arid and less-fertile systems. Increased nitrogen application tends to reduce the efficacy of intensification practices, especially for yields.

3.8. Topsoil SOC changes were associated with increased microbial biomass

Microbial biomass carbon concentrations increased across all four practice combinations with sufficient data. Reduced tillage had the largest effect of any individual practice (Fig. 5a) especially under no-tillage (+35.0%) rather than minimum tillage (+24.1%) (though the confidence intervals overlapped) (Fig. 5b). Intensification (cover cropping and crop rotations) also increased MBC (+17.6%), followed by residue retention (+12.6%). The effect of residue retention on MBC was greater when residue was retained on the soil surface (+30.8%) rather than incorporated into the soil (+14.1%), in contrast with topsoil SOC gains, but the confidence intervals of the two methods overlapped (Fig. 2f; Fig. 5c). Aridity was the most important predictor of the microbial biomass carbon response (Table S13), with the greatest average MBC response in dry sub-humid regions (+25.9%) and the smallest mean response in semi-arid regions (+19.1%) (Table S16). Thus, while microbial biomass carbon concentrations tend to increase across the board, different practices (and within-practice implementation

methods) affect the magnitude of the changes.

3.9. Subsoil SOC responses were more variable and may have offset topsoil SOC gains

While all practices apart from intensification had significantly positive effects on topsoil SOC (Fig. 2b), the subsoil effects were much more variable. We were only able to assess these responses on a subset of our data ($n_{\text{topsoil}} = 2975$ vs $n_{\text{subsoil}} = 498$), and thus part of the variation observed in the subsoil effects may be due to (i) which studies were included and (ii) lower confidence because of smaller sample sizes.

Only residue retention had a significantly positive effect on subsoil SOC stocks (+43.7%), though the estimates varied widely. The remaining practices did not significantly affect subsoil SOC stocks, though the average effect of intensification was slightly negative (−7.3%) and the average effect of reduced tillage on subsoil SOC stocks was substantially negative (−24.3%) (Fig. 6, Table S14).

When we aggregated topsoil and subsoil values to investigate the response of the full soil profile, reduced tillage no longer had a significantly positive effect on SOC stocks ($p > 0.50$), indicating that the increases in the topsoil were offset by decreases in the subsoil. The only individual practice with a significantly positive effect on total SOC stocks was residue retention, where moderate topsoil SOC increases (+12.7%) were bolstered by large average increases in the subsoil (+43.7%) to produce a +20.5% increase across the whole soil profile. Although sample sizes were small, combining residue retention with reduced tillage (RES + RT) significantly increased total SOC stocks (+17.2%), as did combining residue retention with intensification (RES

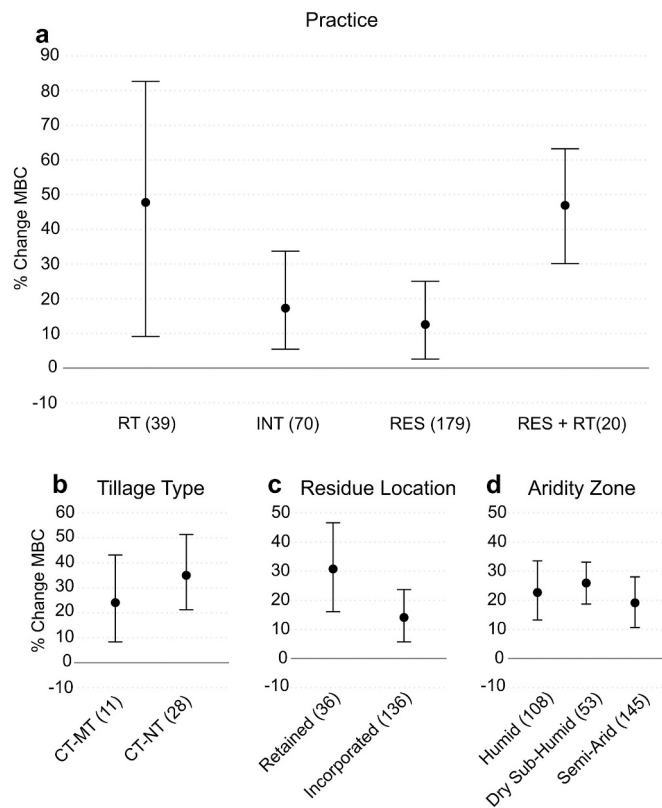


Fig. 5. Microbial biomass carbon (MBC) responses to sustainable practices. Broken down by type of practice used (a), type of tillage in the reduced tillage subset (b), residue location in residue retention subset (c), and aridity zone (d). All analyses based on univariate mixed effects models, and (b) and (c) used the same practice subsetting approach as Fig. 2. Points represent mean estimates and error bars represent bootstrapped 95% confidence intervals, such that if the bars do not overlap the x-axis then the effects are significant. Values in parentheses represent sample sizes. In a, RT = reduced tillage, INT = cropping system intensification (cover cropping and/or crop rotations) and RES = residue retention. CT-MT refers to observations where the conventional practice is conventional tillage and the sustainable practice is minimum tillage; in CT-NT the sustainable practice is no-tillage.

+ INT; +12.9%) (Fig. 6, Table S14). Taken together, while subsoil responses were highly variable, considering subsoil SOC changes produced different responses depending on the practice. Reduced tillage and intensification practices produced negative mean effects on subsoil SOC stocks, which in the case of reduced tillage appeared to offset significant topsoil SOC gains. On the other hand, residue retention increased SOC stocks across the whole soil profile and may alleviate the neutral or negative effects of the other practices.

4. Discussion

4.1. Practices improve both yields and SOC stocks globally

Our global meta-analysis provides clear evidence that practices like reduced tillage, cover cropping, complex crop rotations and residue retention can increase topsoil SOC stocks and/or crop yields and frequently produce win-win outcomes (in 18.1% of observations). In contrast to previous studies (e.g., Vendig et al., 2023), our results suggest that true trade-offs between SOC sequestration and crop yields (where one response is significantly positive while the other is significantly negative) are relatively rare (only 5.7% of observations), although our study assessed a much broader range of management practices. However, our results also highlight that win-wins are not guaranteed, meaning that much can be done to identify the optimal

deployment of practices to maximize SOC sequestration and crop yields.

4.2. Topsoil SOC and crop yield responses are constrained by environmental and agronomic conditions

Across all practices, we observed that the topsoil SOC change was negatively correlated with background (initial) SOC concentrations (Fig. 3b), meaning greater gains were experienced in areas with low baseline carbon contents. Similar results have been reported for individual practices including cover cropping (Vendig et al., 2023) and residue retention (Liu et al., 2014; Xia et al., 2018). Negative correlations between SOC responses and initial SOC could be the result of statistical artifacts (Slessarev et al., 2023), giving us pause in speculating about the actual biogeochemical mechanism. However, we found that the observed relative soil carbon changes (response ratios) were strongly positively correlated with absolute soil carbon stock changes in our data (Fig. S5, see Supplementary Materials for greater details).

Interestingly, there was no such negative relationship between topsoil SOC response and the mineral-associated carbon (MAOC) saturation (Fig. 3c). This indicates either that the soils are not MAOC-saturated (and thus can gain both particulate- and mineral-associated organic carbon) or that the topsoil SOC gains are predominantly the result of particulate organic carbon gains rather than mineral-associated organic carbon gains. Previous studies have found that sustainable agricultural practices have increased carbon storage in both particulate- and mineral-associated pools (Prairie et al., 2023). Thus, it is unclear (at least using global datasets) whether mineral-associated organic carbon saturation can predict potential SOC changes at the global scale.

The SOC response remained constant across a gradient of nitrogen application rates, but the yield response was greatest in low-N systems (Fig. 3d). This echoes previous results (Abdalla et al., 2019; King and Blesh, 2018; Zhao et al., 2022), suggesting that practices like cover cropping can help overcome nitrogen limitation. This overall trend is generally consistent across crop types but is mainly driven by maize and rice systems and less pronounced in wheat (Table S9). Soybean systems respond differently, which is intuitive given their capacity to fix nitrogen, though small sample sizes hamper our ability to draw conclusions about intermediate nitrogen application rates.

Lastly, we observed that both yield and topsoil SOC responses were greater over longer experimental timeframes, though this trend was particularly pronounced for topsoil SOC (Fig. 3e). This is consistent with previous studies showing greater yield and/or SOC effects of cover cropping, reduced tillage (Bai et al., 2019), and residue retention (Xia et al., 2018) over longer experimental durations. While we did not observe the saturation of the SOC response reported by others (Paustian et al., 2019) globally (Fig. 3e) or in a practice-specific context (Fig. 4f), the observations in our dataset had an average study length of 7 years, making extrapolation to longer time scales challenging.

4.3. Heterogenous outcomes of practice implementation generate management trade-offs and constrain win-wins

While true trade-offs (where one outcome responds negatively) may be rare, the magnitude of topsoil SOC and yield responses depended on how and where the practices were implemented.

4.3.1. Reduced tillage

Reduced tillage alone significantly increased topsoil SOC and had no effect on crop yields compared to conventional tillage (Fig. 2b). Within reduced tillage, no-tillage increased topsoil SOC stocks but reduced crop yields while minimum tillage did not have a significant effect on either response (Fig. 2e). Previous research has mostly focused on the effects of no-tillage, though Ogle et al. (2005) similarly found that converting from conventional to minimum tillage had less of an SOC benefit than converting to no-tillage. However, Bai et al. (2019) found that minimum tillage and no-tillage both increased SOC. We sometimes included chisel

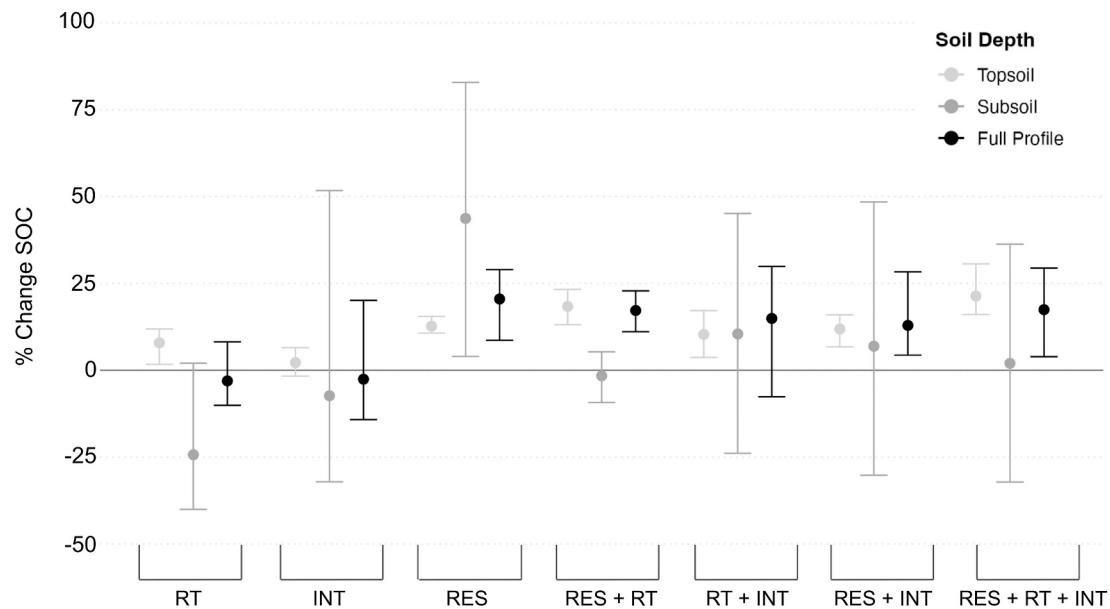


Fig. 6. Soil carbon response by management practice and soil profile. Points represent mean estimates and error bars represent bootstrapped 95% confidence intervals, such that if the bars do not overlap the x-axis then the effects are significant. Topsoil, subsoil and full profile sample sizes are: Reduced tillage (RT: 532, 155, 158), Intensification (cover cropping and/or complex crop rotations) (INT: 494, 41, 44), Residue Retention (RES: 1458, 227, 243), Residue Retention and Reduced Tillage (RES + RT: 246, 43, 41), Reduced Tillage and Intensification (RT + INT: 69, 9, 9), Residue and Intensification (RES + INT: 131, 5, 17), all three practices (RES + RT + INT: 18, 6, 6).

plough as a minimum tillage treatment if the control was mouldboard ploughing, which Bai et al. (2019) did not. Since chisel ploughing is more invasive than other minimum tillage techniques, our neutral result for the effect of minimum tillage on SOC stocks may be conservative and could be alleviated by using less invasive methods. Our result that no-tillage significantly increased topsoil SOC stocks is consistent with the literature consensus (Ogle et al., 2019; Prairie et al., 2023; Shang et al., 2021) and our finding that no-tillage reduced yields (−4.7% across all crops) is consistent with Pittelkow et al. (2015), who showed that no-tillage alone reduced yields by 9.9%. Our results suggest that the decision over which method of reduced tillage is important: no-tillage alone may be optimal for soil carbon sequestration but will sacrifice yields, whereas minimum tillage alone will provide little SOC or yield benefit.

Just as the effects of reduced tillage are impacted by the extent of tillage reduction, they are also modulated by climatic constraints. Across all practices, the greatest SOC response and the lowest yield response was in arid regions (Fig. 3a), though we found that reduced tillage was mostly likely to increase yields and topsoil SOC in warmer (Fig. 4a) regions with low potential evapotranspiration (Fig. 4b) and greater aridity (with an aridity index less than 0.5) (Fig. 4c). In warm and dry climates, reduced tillage can improve soil structure, water infiltration and soil moisture retention (Pittelkow et al., 2015). Our results echo previous studies that have found greater SOC changes in warmer (Ogle et al., 2019; Prairie et al., 2023; Shang et al., 2021) and drier (Sun et al., 2020) climates. Greater SOC changes under reduced tillage in warmer regions (compared to conventional tillage) are likely the result of higher microbial activity in warmer climates that would quickly decompose organic matter in conventionally disturbed fields (Prairie et al., 2023).

4.3.2. Crop rotations and cover cropping

Within cropping system intensification practices, we found that complex crop rotations only slightly increased topsoil SOC stocks (+3.3%) and did not have a significant effect on yields, whereas cover cropping increased both topsoil SOC and crop yields (Fig. 2c). Prairie et al. (2023) similarly showed that cropping intensification increased topsoil SOC, though they reported larger (+12.4%) topsoil organic carbon increases, likely due to their inclusion of perennial cropping

systems and cover cropping under the umbrella of cropping intensification (which would provide greater carbon inputs than most complex annual crop rotations). Unlike Smith et al. (2008), we did not find that complex crop rotations increased crop yields, though in our study a complex crop rotation could include the transition from a maize monoculture to a maize-wheat rotation, a relatively modest increase in complexity that might dilute potential yield benefits of more complex rotations. This suggests that the benefits of complex crop rotations in annual cropping systems are limited unless the complex crop rotation includes a crop that provides either additional carbon or nitrogen inputs or is grown during the fallow period (i.e., cover cropping).

Our results demonstrated that cover cropping increased both topsoil SOC and crop yields (Fig. 2c), though these effects were heterogeneous across cover crop functional groups, generating a management trade-off. In this case, grass cover crops led to slightly greater SOC increases but the yield benefits were non-significant. Legume cover crops led to significant yield increases but smaller SOC gains compared to grass species. Our results echoed previous studies showing that cover cropping increased topsoil SOC (Fig. 2c) (McClelland et al., 2021; Poeplau and Don, 2015). Previous studies (Abdalla et al., 2019; McClelland et al., 2021; Poeplau and Don, 2015) found that functional group (typically considered to be legume versus non-legume) had no effect on the SOC response. However, these studies did not distinguish between different types of non-legume cover crops; we found that using a grass cover crop produced a slightly higher SOC response than a legume cover crop. Conversely, Jian et al. (2020) found greater SOC increases with legume cover crops compared to grass cover crops. One explanation for the discrepancies across studies could be that functional group inadequately captures key species-specific traits like biomass C:N ratio which may better predict SOC responses.

Our finding that cover cropping increased crop yields by an average of 26.6% (Fig. 2c), particularly with legume cover crops (Fig. 2d) and in low-N systems (Fig. 4h), aligns with several studies. Vending et al. (2023) showed that cover cropping increased yields by 10.9%, with legumes in low-N fertilized systems leading to even larger yield increases. Similarly, Zhao et al. (2022) showed that legume pre-crops increased yields by 20.4%. However, results are context-dependent: Lobell et al. (2025)

found yield declines of 3% for maize and 2% for soybean. Unlike Lobell et al. (2025), our data comes from experimental field trials with standardized management protocols; thus, we are unable to account for the effects of delayed planting that they suggest was the main driver of yield declines. Moreover, Lobell et al. (2025) focused on maize and soybean production in the U.S., systems characterized by high-input agriculture with high nitrogen application rates. In our study, the average nitrogen application rate overall was 125 kg N/ha, and the average for cover cropping studies was only 85 kg N/ha, with over one-third of cover cropping observations being unfertilized.

We observed (as have many other studies) that the yield effects of cover cropping were greatest in systems with low nitrogen application rates (Abdalla et al., 2019; Vendig et al., 2023), suggesting that cover cropping helps overcome nitrogen limitation, but this effect is masked when sufficient external nitrogen is supplied. This may explain why nitrogen-fixing legume cover crops significantly increased crop yields, but grass and brassica species did not (Fig. 2d). Again, this varied by crop type, with rice and maize showing positive yield responses below 160 kg N/ha, wheat below 100 kg N/ha, and soybean never showing significantly positive responses across the nitrogen gradient (though only becoming significantly negative above 160 kg N/ha; Fig. S5), indicative of crop-specific nitrogen requirements constraining cover crop effectiveness.

4.3.3. Residue retention

The consistent and strong benefits of residue retention for yields and topsoil SOC, illustrated in both our study and past work (Fig. 2b, f) (Liu et al., 2014; Shang et al., 2021; Xia et al., 2018), suggests that residue retention is the individual practice that will most consistently produce win-wins; in contrast, reduced tillage and intensification practices may be more likely to generate trade-offs. However, the method of retention was important: surface residue retention did not significantly affect topsoil SOC stocks. This is reflective of the relatively small amount of transfer from aboveground biomass to SOC in croplands: on average, 8.3% of aboveground carbon inputs are retained in soil organic matter whereas 46% of belowground carbon inputs are retained (Jackson et al., 2017). Incorporation of residues into the soil facilitates the stabilization of carbon inputs within soil aggregates and protection from decomposition.

There is no clear consensus concerning the climatic constraints on residue retention. A positive relationship between mean annual temperature and SOC response would be expected for both residue retention and cover cropping – and indeed we observed it for cover cropping (Fig. 4g) – likely because residue and cover crop-derived carbon inputs compensate for faster organic matter turnover in warm conditions (Shang et al., 2021). Contrary to expectations but consistent with other studies (Liu et al., 2014) we found no significant relationship between mean annual temperature and SOC response to residue retention, though we found that the yield response was greatest in warmer climates. Interestingly, we found that potential evapotranspiration had a positive relationship with SOC response (Fig. 4e) (Table S11) but a negative relationship with the yield response. This indicates that residue retention is likely to produce win-wins for soil carbon sequestration and crop productivity in less water-stressed environments (low PET) while the yield effect becomes insignificant in more water-stressed environments (high PET). Our findings add to the discussion by suggesting that all three main practices (reduced tillage, residue and cover cropping) are most likely to produce win-wins in warmer climates, though the complexities of climatic constraints (temperature, aridity, water demand, etc.) and interactions with other important predictors make it challenging to generalize using climatic variables alone.

4.4. Crop type and practice interactions generate further management trade-offs

Our results show that crop type was an important determinant of the

yield response, indicating that gains in maize and wheat yields are more common than rice and soybeans. Consequently, matching practice selection with crop type reveals further management trade-offs. For example, in both maize and wheat systems, residue retention led to the largest average topsoil SOC and yield gains of the individual practices (Table 1), as shown by Shang et al. (2021). While in rice and soybean systems we found that residue retention led to the largest average SOC gains, intensification led to the largest average yield gains of the individual practices (Table 1). Thus, when choosing between individual practices, rice and soybean farmers may face a trade-off between maximizing yields (via intensification) or maximizing soil carbon sequestration (via residue retention).

4.5. Practice stacking may alleviate management trade-offs associated with individual practices

Consistent with our expectations and the literature (McClelland et al., 2021; Pittelkow et al., 2015), we found that combining multiple practices tends to have a beneficial effect on both topsoil SOC and yield outcomes. For example, combining reduced tillage or residue retention with intensification tended to increase the average yield response, and combining reduced tillage or intensification with residue retention helped overcome the negligible impacts of intensification and reduced tillage on SOC and yields, respectively. Previous meta-analyses have shown the benefits of practice stacking, such as improving the SOC benefit of cover cropping (McClelland et al., 2021). This may be context-specific however, as cover crop effectiveness is limited in dry climates due to water competition with main crops (Bai et al., 2019; D. B. Lobell et al., 2025). This suggests there may be limits to practice stacking, as combining other practices with cover cropping may provide no additional benefit or even be detrimental in arid climates.

Pittelkow et al. (2015) showed that stacking no-tillage with residue retention or crop rotations (or both) reduced the negative impact of no-tillage on yields, though the responses were still negative. In contrast, our results suggest that stacking reduced tillage with residue retention, for example, can lead to significant yield increases (Fig. 2b). However, Pittelkow et al. (2015) compared no-tillage to conventional tillage where residue management and crop rotations were consistent across treatments (i.e., they just assessed the impact of different baseline residue management and/or crop rotations on the effectiveness of no-tillage), whereas our analysis examines the stacking of additional practices. Still, we similarly conclude that residue retention and cropping intensification should be considered the cornerstone conservation agriculture practices rather than reduced tillage, though reduced tillage can still provide other economic and environmental co-benefits.

Few if any previous studies have assessed how the effects practice stacking vary by crop type and the extent to which stacking practices helps to maximize SOC and yield outcomes. By assessing the effects by crop type in our study, we found important management trade-offs (Table S8). For example, in both rice and maize systems, the yield gain is maximized by combining residue retention with intensification, whereas the average topsoil SOC gain is maximized by combining residue retention with reduced tillage. Thus, practice stacking is crucial not only to alleviating negative or neutral impacts of individual practices but also help to maximize crop-specific SOC and yield outcomes.

4.6. Topsoil SOC changes are associated with increases in microbial biomass

Overall, sustainable agricultural practices increased microbial biomass carbon (MBC) (Fig. 5a). These results are broadly consistent with the literature (Li et al., 2018; Zuber and Villamil, 2016) and likely reflect increased carbon inputs and soil microclimate conditions that favour microbial growth (Li et al., 2018; Liu et al., 2014; Prairie et al., 2023). However, the implications for SOC dynamics differ between practices: while residue retention and cropping intensification may

increase topsoil SOC stocks partly through enhanced microbial activity that converts carbon inputs into soil organic matter (Liu et al., 2014; McClelland et al., 2021), reduced tillage may increase topsoil SOC primarily by reducing decomposition and maintaining soil structure (Shang et al., 2021), suggesting the concurrent increase in MBC could potentially constrain rather than accelerate topsoil SOC sequestration when reducing tillage.

4.7. Subsoil SOC responses are highly variable and may offset topsoil SOC gains

While subsoil carbon effects were highly variable (Fig. 6), they adjusted our understanding of practice effectiveness beyond simply reducing statistical confidence. For reduced tillage, the inclusion of subsoil data transformed what appeared to be a significantly positive topsoil effect into a non-significant whole-profile response, indicating that subsoil losses likely offset topsoil gains. Similarly, cropping intensification showed no significant effect on SOC stocks across the full soil profile. This is consistent with previous research showing neutral or negative effects of some sustainable practices on subsoil carbon stocks (e.g., Ogle et al., 2019; Prairie et al., 2023). In contrast, residue retention maintained significant positive effects when subsoil responses were included, with significant subsoil gains bolstering topsoil increases to produce substantial carbon sequestration (+20.5%) across the whole profile. Thus, while residue retention consistently increases SOC stocks regardless of depth, subsoil responses may offset topsoil gains resulting from other practices such as reduced tillage.

4.8. Uncertainties in net emissions limit accurate estimates of climate benefits

We assessed soil carbon changes relative to the baseline of conventional agricultural management practices, not relative to the baseline ecosystem state, meaning the “gains” we observed may simply represent slower or reduced losses of soil carbon after land conversion. Still, given global food production needs and the fact that conservation agriculture practices cover a minority of global agricultural land, our use of the conventional practice as the baseline is appropriate. Beyond soil carbon responses, the effects of these practices on other on-farm and downstream greenhouse gas emissions are important uncertainties. For example, Qiu et al. (2024) found that cover cropping increased yields by 2.33% and SOC stocks by 6.46% but also increased N₂O and CH₄ emissions by 29.5% and 42.3%, respectively. Similarly, there is evidence that residue retention may increase CH₄ emissions in rice paddies and N₂O emissions in upland farms, which could offset carbon sequestration in soils (Shang et al., 2021). Paustian et al. (2016) suggested that increases in microbial biomass carbon (and microbial activity) like those we observed (Fig. 5) may drive increased CH₄ and N₂O emissions. However, sustainable practices could also reduce on-farm emissions by reducing fuel emissions (in the case of reduced tillage) or fertilizer-related emissions (by growing N-fixing cover crops or recycling nutrients through residue retention). As such, the net effect of these practices on climate change mitigation remains uncertain and is of vital importance for future research to enable effective implementation at the global scale.

5. Conclusions

Our meta-analysis showed that, on a global scale, sustainable agricultural practices generally increased topsoil SOC stocks and crop yields, though the magnitude and significance of the change depended on the type of practice, crop, and the practice implementation method. By considering the interactions between crop types and multiple practices thought to bolster SOC, we reveal that management trade-offs primarily arise from determining which practice in which crop will maximize SOC and yield outcomes. For example, in rice systems, residue retention

maximized topsoil SOC gains while cropping intensification maximized yield gains, though such trade-offs may be alleviated by combining multiple practices. Across the practices, win-wins were more likely in water- and nitrogen-limited systems over longer timeframes, indicating there is substantial global potential for deployment. However, uncertainties around subsoil SOC responses as well as on-farm and downstream emissions must be addressed. While we assessed relative yield and SOC changes, future work should determine the spatially explicit absolute changes in soil carbon stocks, total food production, knock-on emissions, and economic implications of sustainable agricultural practices to ensure effective policy design and large-scale implementation of sustainable practices.

CRediT authorship contribution statement

David G. Encarnation: Writing – review & editing, Writing – original draft, Visualization, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Robert S. Powell:** Writing – review & editing, Software, Conceptualization. **Pete Smith:** Writing – review & editing, Conceptualization. **Adam F.A. Pellegrini:** Writing – review & editing, Writing – original draft, Supervision, Software, Project administration, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2026.181438>.

Data availability

Data were collected from published materials that are freely accessible. Please see Table S3 for sources of spatial datasets used in our study. All the empirical data that support the main findings of this study as well as the R code used in the analyses presented in this paper will be made available in GitHub following acceptance and can be accessed at https://github.com/davidencarnation/sustainable_ag_SOC_yield_meta_analysis.

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